Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

A wireless sensor needs to communicate to the base station the state of the processes running on its CPU. Each process can be either in run (R), wait (W) or sleep (S) state. Figure 1 illustrate the transition probabilities $p(s_t|s_{t-1})$. The sampling frequency is 10 Hz. The marginal probabilities can be determined to be

$$p(s_t = R) = 0.2, \quad p(s_t = W) = 0.2, \quad p(s_t = S) = 0.6$$

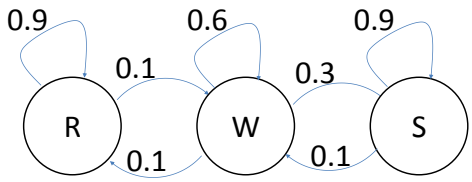


Figure 1: Each node represent a state, each edge a state transition with the associated probability.

- Compute the entropy of the source X (in bits/symbol) that emits symbols from the alphabet $\{R, W, S\}$, assuming that $p(s_t|s_{t-1}) = p(s_t)$ (discrete memoryless source).
- Simulate the encoding and decoding of the sequence $SSWR$ using Elias coding in three cases:
  - Assuming that $p(s_t|s_{t-1}) = p(s_t)$ (discrete memoryless source).
  - Assuming the knowledge of the conditional probabilities as in Figure 1.
  - Assuming that the source statistics are unknown and they are determined at encoding time as symbols are observed.

  Compare the lengths of the associated codeword in the three cases.
- When the sensor is in running state, it communicates its reading, quantizing it with $R = 4$ bits/symbol. Compute the overall data rate (to communicate both the sequence of states and the readings) when source memory is not exploited and there are $m = 10$ processes.

2 Question - 5 pts

Illustrate the design of the Lloyd-Max quantizer. Compare the rate-distortion performance of the Lloyd-Max quantizer with other possible quantizer designs.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote the source of Exercise 1. Assume X be a discrete memoryless source
   - [T] [F] $H(X) = 0.4127$ bits/symbol
   - [T] [F] $i(W) \geq i(S)$, where $i(\cdot)$ is the information content of a symbol
   - [T] [F] The average code length of Shannon coding groups of $N = 3$ symbols is less than 1.7043 bits/symbol
   - [T] [F] $l(W) \leq l(S)$, where $l(\cdot)$ is the codeword length adopted by Huffman coding
2. Consider a speech coding system
   - [T] [F] The long term predictor is useful when coding unvoiced speech
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder
   - [T] [F] The number of poles of the whitening filter $A(z)$ depends on the number of formants below 4 kHz
3. Consider a block-based image coding system
   - [T] [F] With threshold quantization, each block is allocated the same bit budget
   - [T] [F] Increasing the block size, the transform coding gain increases
   - [T] [F] DPCM can be useful to exploit inter-block correlations.

4 Question - 12 pts

Load the image by means of the Matlab command `imread('rice.png')`. Let X denote a source that emits the pixel values of the image from the alphabet $\{0, 1, \ldots, 255\}$.

- Compute the empirical entropy of the source X, assuming that symbols are statistically independent
- Compute the empirical entropy of the source X, conditioned on the knowledge of the value of the pixel located either above or at the left of the current pixel
- Implement a DPCM codec that exploits the best between the two predictors and plot the rate-distortion curve in a PSNR (dB) -rate (bpp) chart.